

8. For each of these generating functions, provide a closed formula for the sequence it determines.

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|---------------------------|--------------------------|
| a) $(x^2 + 1)^3$ | b) $(3x - 1)^3$ |
| c) $1/(1 - 2x^2)$ | d) $x^2/(1 - x)^3$ |
| e) $x - 1 + (1/(1 - 3x))$ | f) $(1 + x^3)/(1 + x)^3$ |
| *g) $x/(1 + x + x^2)$ | h) $e^{3x^2} - 1$ |

16. Use generating functions to find the number of ways to choose a dozen bagels from three varieties—egg, salty, and plain—if at least two bagels of each kind but no more than three salty bagels are chosen.

24. a) What is the generating function for $\{a_k\}$, where a_k is the number of solutions of $x_1 + x_2 + x_3 + x_4 = k$ when x_1, x_2, x_3 , and x_4 are integers with $x_1 \geq 3, 1 \leq x_2 \leq 5, 0 \leq x_3 \leq 4$, and $x_4 \geq 1$?
- b) Use your answer to part (a) to find a_7 .

38. Use generating functions to solve the recurrence relation $a_k = a_{k-1} + 2a_{k-2} + 2^k$ with initial conditions $a_0 = 4$ and $a_1 = 12$.